

Matrix Methods for Linear Systems

9.1 Introduction

In this chapter we return to the analysis of systems of differential equations. When the equations in the system are *linear*, matrix algebra provides a compact notation for expressing the system. In fact, the notation itself suggests new and elegant ways of characterizing the solution properties, as well as novel, efficient techniques for explicitly obtaining solutions.

In Chapter 5 we analyzed physical situations wherein two fluid tanks containing brine solutions were interconnected and pumped so as ultimately to deplete the salt content in each tank. By accounting for the various influxes and outfluxes of brine, a system of differential equations for the salt contents ($x(t)$ and $y(t)$) of each tank was derived; a typical model is

$$(1) \quad \begin{aligned} dx/dt &= -4x + 2y, \\ dy/dt &= 4x - 4y. \end{aligned}$$

Express this system in matrix notation as a single equation.

The right-hand side of the first member of (1) possesses a mathematical structure that is familiar from vector calculus; namely, it is the **dot product**[†] of two vectors:

$$(2) \quad -4x + 2y = \begin{bmatrix} -4 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}.$$

Similarly, the second right-hand side in (1) is the dot product

$$4x - 4y = \begin{bmatrix} 4 & -4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}.$$

The frequent occurrence in mathematics of arrays of dot products, such as evidenced in the system (1), led to the development of **matrix algebra**, a mathematical discipline whose basic operation—the matrix product—is the arrangement of a set of dot products according to the following plan:

$$\begin{bmatrix} -4 & 2 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} -4 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} \\ \begin{bmatrix} 4 & -4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} \end{bmatrix} = \begin{bmatrix} -4x + 2y \\ 4x - 4y \end{bmatrix}.$$

In general, the product of a **matrix**—i.e., an m by n rectangular array of numbers—and a **column vector** is defined to be the collection of dot products of the *rows* of the matrix with the

[†]Recall that the dot product of two vectors $\mathbf{u}$ and $\mathbf{v}$ equals the length of $\mathbf{u}$ times the length of $\mathbf{v}$ times the cosine of the angle between $\mathbf{u}$ and $\mathbf{v}$. However, it is more conveniently computed from the *components* of $\mathbf{u}$ and $\mathbf{v}$ by the “inner product” indicated in equation (2).

vector, arranged as a column vector:

$$\begin{bmatrix} \text{row \# 1} \\ \text{row \# 2} \\ \vdots \\ \text{row \# m} \end{bmatrix} \begin{bmatrix} \mathbf{v} \end{bmatrix} = \begin{bmatrix} [\text{row \# 1}] \cdot \mathbf{v} \\ [\text{row \# 2}] \cdot \mathbf{v} \\ \vdots \\ [\text{row \# m}] \cdot \mathbf{v} \end{bmatrix},$$

where the vector $\mathbf{v}$ has n components; the dot product of two n -dimensional vectors is computed in the obvious way:

$$[a_1 \ a_2 \ \cdots \ a_n] \cdot [x_1 \ x_2 \ \cdots \ x_n] = a_1x_1 + a_2x_2 + \cdots + a_nx_n.$$

Using the notation for the matrix product, we can write the system (1) for the interconnected tanks as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -4 & 2 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

The following example demonstrates a four-dimensional implementation of this notation. Note that the coefficients in the linear system need not be constants.

Example 1 Express the system

$$\begin{aligned} x'_1 &= 2x_1 + t^2x_2 + (4t + e^t)x_4, \\ x'_2 &= (\sin t)x_2 + (\cos t)x_3, \\ x'_3 &= x_1 + x_2 + x_3 + x_4, \\ x'_4 &= 0 \end{aligned} \quad (3)$$

as a matrix equation.

Solution We express the right-hand side of the first member of (3) as the dot product

$$2x_1 + t^2x_2 + (4t + e^t)x_4 = [2 \ t^2 \ 0 \ (4t + e^t)] \cdot [x_1 \ x_2 \ x_3 \ x_4].$$

The other dot products are similarly identified, and the matrix form is given by

$$\begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \\ x'_4 \end{bmatrix} = \begin{bmatrix} 2 & t^2 & 0 & (4t + e^t) \\ 0 & \sin t & \cos t & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}. \quad \blacklozenge$$

In general, if a system of differential equations is expressed as

$$\begin{aligned} x'_1 &= a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n \\ x'_2 &= a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n \\ &\vdots \\ x'_n &= a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n, \end{aligned}$$

it is said to be a linear homogeneous system in **normal form**.[†] The matrix formulation of such

[†]The normal form was defined for general systems in Section 5.3, page 256.

a system is then

$$\mathbf{x}' = \mathbf{A}\mathbf{x},$$

where $\mathbf{A}$ is the **coefficient matrix**

$$\mathbf{A} = \mathbf{A}(t) = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{bmatrix}$$

and $\mathbf{x}$ is the solution vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Note that we have used $\mathbf{x}'$ to denote the vector of derivatives

$$\mathbf{x}' = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}' = \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix}.$$

Example 2 Express the differential equation for the undamped, unforced mass–spring oscillator (recall Section 4.1, page 152)

$$(4) \quad my'' + ky = 0$$

as an equivalent system of first-order equations in normal form, expressed in matrix notation.

Solution We have to express the *second* derivative, y'' , as a *first* derivative in order to formulate (4) as a first-order system. This is easy; the acceleration y'' is the derivative of the *velocity* $v = y'$, so (4) becomes

$$(5) \quad mv' + ky = 0.$$

The first-order system is then assembled by identifying v with y' , and appending it to (5):

$$\begin{aligned} y' &= v \\ mv' &= -ky. \end{aligned}$$

To put this system in normal form and express it as a matrix equation, we need to divide the second equation by the mass m :

$$\begin{bmatrix} y \\ v \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -k/m & 0 \end{bmatrix} \begin{bmatrix} y \\ v \end{bmatrix}. \quad \blacklozenge$$

In general, the customary way to write an n th-order linear homogeneous differential equation

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = 0$$

as an equivalent system in normal form is to define the first $(n - 1)$ derivatives of y (including y , the zeroth derivative, itself) to be new unknowns:

$$\begin{aligned}x_1(t) &= y(t), \\x_2(t) &= y'(t), \\&\vdots \\x_n(t) &= y^{(n-1)}(t).\end{aligned}$$

Then the system consists of the identification of $x_j(t)$ as the derivative of $x_{j-1}(t)$, together with the original differential equation expressed in these variables (and divided by $a_n(t)$):

$$\begin{aligned}x_1' &= x_2, \\x_2' &= x_3, \\&\vdots \\x_{n-1}' &= x_n, \\x_n' &= -\frac{a_0(t)}{a_n(t)}x_1 - \frac{a_1(t)}{a_n(t)}x_2 - \cdots - \frac{a_{n-1}(t)}{a_n(t)}x_n.\end{aligned}$$

For systems of two or more higher-order differential equations, the same procedure is applied to each unknown function in turn; an example will make this clear.

Example 3 The coupled mass–spring oscillator depicted in Figure 5.26 on page 283 was shown to be governed by the system

$$\begin{aligned}(6) \quad 2\frac{d^2x}{dt^2} + 6x - 2y &= 0, \\ \frac{d^2y}{dt^2} + 2y - 2x &= 0.\end{aligned}$$

Write (6) in matrix notation.

Solution We introduce notation for the lower-order derivatives:

$$(7) \quad x_1 = x, \quad x_2 = x', \quad x_3 = y, \quad x_4 = y'.$$

In these variables, the system (6) states

$$\begin{aligned}(8) \quad 2x_2' + 6x_1 - 2x_3 &= 0, \\ x_4' + 2x_3 - 2x_1 &= 0.\end{aligned}$$

The normal form is then

$$\begin{aligned}x_1' &= x_2, \\x_2' &= -3x_1 + x_3, \\x_3' &= x_4, \\x_4' &= 2x_1 - 2x_3\end{aligned}$$

or in matrix notation

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 0 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$

9.1 EXERCISES

In Problems 1–6, express the given system of differential equations in matrix notation.

1. $x' = 7x + 2y$,
 $y' = 3x - 2y$
2. $x' = y$,
 $y' = -x$
3. $x' = x + y + z$,
 $y' = 2z - x$,
 $z' = 4y$
4. $x'_1 = x_1 - x_2 + x_3 - x_4$,
 $x'_2 = x_1 + x_4$,
 $x'_3 = \sqrt{\pi}x_1 - x_3$,
 $x'_4 = 0$
5. $x' = (\sin t)x + e^t y$,
 $y' = (\cos t)x + (a + bt^3)y$
6. $x'_1 = (\cos 2t)x_1$,
 $x'_2 = (\sin 2t)x_2$,
 $x'_3 = x_1 - x_2$

In Problems 7–10, express the given higher-order differential equation as a matrix system in normal form.

7. The damped mass–spring oscillator equation $my'' + by' + ky = 0$
8. Legendre's equation $(1 - t^2)y'' - 2ty' + 2y = 0$
9. The Airy equation $y'' - ty = 0$
10. Bessel's equation $y'' + \frac{1}{t}y' + \left(1 - \frac{n^2}{t^2}\right)y = 0$

In Problems 11–13, express the given system of higher-order differential equations as a matrix system in normal form.

11. $x'' + 3x + 2y = 0$,
 $y'' - 2x = 0$
12. $x'' + 3x' - y' + 2y = 0$,
 $y'' + x' + 3y' + y = 0$
13. $x'' - 3x' + t^2y - (\cos t)x = 0$,
 $y''' + y'' - tx' + y' + e^t x = 0$

9.2 Review 1: Linear Algebraic Equations

Here and in the next section we review some basic facts concerning linear algebraic systems and matrix algebra that will be useful in solving linear systems of differential equations in normal form. Readers competent in these areas may proceed to Section 9.4.

A set of equations of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

(where the a_{ij} 's and b_i 's are given constants) is called a *linear system of n algebraic equations* in the n unknowns $x_1, x_2, \dots, x_n$. The procedure for solving the system using elimination methods is well known. Herein we describe a particularly convenient implementation of the method called the *Gauss–Jordan elimination algorithm*.[†] The basic idea of this formulation is to use the first equation to eliminate x_1 in all the other equations; then use the second equation to eliminate x_2 in all the others; and so on. If all goes well, the resulting system will be “uncoupled,” and the values of the unknowns $x_1, x_2, \dots, x_n$ will be apparent. A short example will make this clear.

[†]The Gauss–Jordan algorithm is neither the fastest nor the most accurate computer algorithm for solving a linear system of algebraic equations, but for solutions executed by hand it has many pedagogical advantages. Usually it is much faster than *Cramer's rule*, described in Appendix D.